

# What Spatial Fine-Tuning Actually Changes: A Multi-Seed Decomposition of Accuracy, Evidence Dependence, and Semantic Consistency in Vision–Language Models

Anonymous WACV [Datasets Track](#) submission

Paper ID \*\*\*\*\*

## Abstract

001 *Fine-tuning a vision–language model (VLM) on a spatial*  
002 *reasoning task usually improves benchmark accuracy. But*  
003 *“better accuracy” is not a single capability: the same adap-*  
004 *tation may change how much the model depends on the*  
005 *correct image, how coherently it applies relational seman-*  
006 *tics, and how it responds to a causal image transformation,*  
007 *in different directions and at different rates. We treat or-*  
008 *ordinary LoRA fine-tuning on the Visual Spatial Reasoning*  
009 *(VSR) task as a controlled intervention and measure, per*  
010 *training seed and across two backbone families, a vector*  
011 *of capability changes: benchmark gain  $\Delta A$ , correct-image*  
012 *dependence  $\Delta G$  (the normal-minus-shuffle accuracy gap,*  
013 *relative to zero-shot), and semantic pair consistency  $\Delta C$*   
014 *(relcomp/facing-antonym flip-law compliance), together with*  
015 *Tier-C transformation behavior under global horizontal re-*  
016 *flexion (transformed accuracy, pair consistency, joint cor-*  
017 *rectness). Across 3 fresh training seeds per family (plus the*  
018 *committed seed-0),  $\Delta A$  is positive for every seed on both*  
019 *backbones (7B:  $+0.0506 \pm 0.0046$ ; 2B:  $+0.0316 \pm 0.0009$ )*  
020 *and  $\Delta G$  is positive for every seed (7B:  $+0.0440 \pm 0.0054$ ;*  
021 *2B:  $+0.0316 \pm 0.0026$ ). Fresh-seed semantic pair consis-*  
022 *tency stays within 0.05 of the legacy General checkpoint*  
023 *in both families, and all fresh 2B answer-update rates ex-*  
024 *ceed the 2B zero-shot rate. A post-confirmatory exten-*  
025 *sion on the contemporary Qwen3-VL-8B backbone repro-*  
026 *duces the primary  $\Delta A/\Delta G$  pattern ( $+0.032/+0.036$ ) and*  
027 *the transformed-accuracy gain ( $+0.045$ ). We conclude that*  
028 *ordinary spatial fine-tuning is not a scalar capability update:*  
029 *benchmark accuracy, correct-image dependence, and seman-*  
030 *tic consistency can be described—and must be reported—as*  
031 *a jointly measured vector whose components replicate across*  
032 *seeds and families.*

## 1. Introduction

Spatial reasoning is central to multimodal systems that operate in physical environments: a useful VLM must determine whether one object is to the left of another, above it, in front of it, or facing it, and it must do so using the image—not a language prior. A growing line of work shows that benchmark accuracy can diverge from genuine visual grounding: models can answer correctly while depending weakly on the image [2, 6, 7], and consistency across conditions can coexist with error [1]. Prior work has also introduced grounding-aware training objectives that explicitly reward visual dependence [5].

What has not been examined is whether *ordinary* spatial task fine-tuning—the standard LoRA recipe with no grounding-aware objective—moves benchmark accuracy, correct-image dependence, and semantic pair consistency as a jointly measured vector of capability changes, and whether those changes replicate across training seeds and backbone families. This is the gap we address.

We use Visual Spatial Reasoning (VSR) [3] as the task. VSR contains natural-image statements about object-relative relations (left/right, above/below, facing, containment, and others) paired with true/false labels. Following the frozen protocol of the underlying study (documented in the repository decision log), we evaluate every checkpoint under a corrected battery of six conditions: *normal* (the frozen prompt and image), *shuffle* (frozen derangement of the image), *relcomp* (strict complement pairs: language changes, expected label flips), *facingcomp* (facing-antonym flip-law diagnostic), and the Tier-C pair *hflip\_flip* / *hflip\_invariant* (global horizontal reflection; language fixed; left/right truth must flip, vertical/depth truth must stay).

We train three fresh General-LoRA seeds per backbone (7B Qwen2-VL-7B and 2B SmolVLM2-2B) using the identical recipe as the committed seed-0 run, then measure, per seed:

- $\Delta A$  (**benchmark gain**): normal accuracy after tuning minus zero-shot normal accuracy;

- $\Delta G$  (**correct-image dependence**): the change in the normal-minus-shuffle accuracy gap relative to zero-shot;
- $\Delta C$  (**semantic pair consistency**): relcomp/facingcomp flip-law compliance and its seed stability;
- **Tier-C transformation behavior**: three separate quantities under global reflection—transformed accuracy ( $A_{\text{transform}}$ ), pair consistency ( $C_{\text{pair}}$ ), and joint correctness (both\_correct)—following the frozen metric definitions of the underlying study.

Every headline number in this paper was recomputed independently from raw prediction artifacts by a standalone audit script (`scripts/audit_paper2_numbers_independent.py`) explicitly that imports no analyzer module; the audit passed with zero discrepancies and zero failed claims. We report means and standard deviations across the three fresh seeds; seeds are independent training draws, not ordered stages.

Our primary finding is that the adaptation decomposes cleanly and reproducibly.  $\Delta A$  is positive for all six fresh seeds across both families;  $\Delta G$  is positive for all six; semantic pair consistency stays within 0.05 of the legacy General checkpoint for every fresh seed; and all fresh 2B Tier-C answer-update rates exceed the 2B zero-shot rate. A post-confirmatory extension on the contemporary Qwen3-VL-8B backbone reproduces the primary  $\Delta A/\Delta G$  pattern and the transformed-accuracy gain, and is explicitly labeled exploratory rather than preregistered.

We make three contributions:

- **A controlled multi-seed, cross-backbone decomposition of ordinary spatial fine-tuning** into benchmark gain, correct-image dependence, and semantic pair consistency, with all quantities recomputed independently and frozen before manuscript mode.
- **A seed-robust result set**: every component of the vector replicates across three fresh training seeds on two backbone families, closing the single-checkpoint-anecdote criticism.
- **A scoped contemporary-backbone extension** (Qwen3-VL-8B) that reproduces the primary adaptation pattern, honestly labeled as post-confirmatory external validation.

## 2. Related Work

**Accuracy versus visual grounding.** A central and rapidly growing observation is that benchmark accuracy can overstate genuine visual dependence. *Beyond Accuracy* (Zafar et al., 2026) [6] evaluates medical VLMs with real, blank, and shuffled images and defines the Visual Reliance Score (VRS) as  $A(\text{real}) - A(\text{shuffle})$ ; they find that RLVR can improve accuracy while degrading visual grounding. Our *correct-image dependence* (the normal-minus-shuffle gap,  $G$ ) is the same construction; we do not claim it as novel, and we report the zero-shot-relative change  $\Delta G$  as the adaptation quantity. *Seeing without Looking* (Lan et al., 2026) [2] applies global

and localized visual interventions to show that predictions can remain robust after evidence removal; it audits pretrained models rather than a fine-tuning transition. *VisualFLIP* (Zhu et al., 2026) [7] introduces a paired benchmark where the question stays fixed and a minimal task-critical edit deterministically flips the gold answer, with pair accuracy and Collapse Rate as metrics; its dataset and harness are not yet publicly released, and we do not claim to reproduce its protocol. Our Tier-C horizontal reflection is a global image intervention, not a minimal local edit; we report our pair metrics as *collapse-style paired answer-update metrics following VisualFLIP*, with the intervention difference stated

**Consistency versus grounding.** *Consistent Yet Wrong* (Bhat and Yamasaki, 2026) [1] introduces ViewDiag and shows that multi-view consistency can coexist with substantial error, indicating prior-driven collapse rather than evidence-sensitive reasoning. Their dissociation between consistency and grounding is the static analogue of our  $\Delta C$ -versus- $\Delta G$  decomposition, but they do not study a fine-tuning transition. *VisualFLIP*'s collapse rate is a similar paired-consistency notion for minimal-edit pairs.

**Grounding-aware training objectives.** *CORAL* (Zafar et al., 2026) [5] fine-tunes Qwen2.5-VL-7B with a Contrastive Grounding Objective that penalizes answer invariance under hard-negative image swaps, improving accuracy and reducing visual hallucination in medical VQA. *Med-OPD* (Qian et al., 2026) [4] adds evidence-aware on-policy distillation for medical VLMs. These works *design* grounding-aware objectives; we study what *ordinary* fine-tuning already changes, without modifying the objective.

**Positioning.** No prior work, to our knowledge, treats ordinary spatial task fine-tuning as a controlled intervention and measures  $\Delta A$ ,  $\Delta G$ , and  $\Delta C$  (plus Tier-C transformation behavior) jointly, per training seed, across two backbone families, with a contemporary-backbone extension. Our novelty claim is precisely this joint, seed-robust decomposition—not the normal-minus-shuffle gap, not “accuracy differs from grounding,” and not a new training objective.

## 3. Protocol

**Task and data.** We use Visual Spatial Reasoning (VSR) [3], test split ( $n = 2195$  rows). Statements are true/false sentences about object-relative relations; labels are frozen. The frozen prompt template, generation settings (greedy, `max_new_tokens = 5`), image preprocessing (uniform 392 px long-side cap), and cache keys (md5 of the image URL) are identical to the underlying committed protocol (documented in the repository decision log and protocol

freeze documents).

**Evaluation battery (corrected, frozen).** Following the battery-drift correction in the decision log, the seed campaign is evaluated with the already-committed legacy grounding drivers, not the drifted heavy battery. The corrected battery has six conditions with canonical frozen subset sizes:

| condition                                               | $n$  |
|---------------------------------------------------------|------|
| normal / shuffle / blank / text_only                    | 2195 |
| relcomp (strict complements)                            | 666  |
| facingcomp (facing-antonym)                             | 103  |
| hflip_flip (global reflection, flip-expected)           | 245  |
| hflip_invariant (global reflection, expected-invariant) | 421  |

The shuffle condition uses the frozen cycle-derangement mapping (seed 20260810); identity and eligibility come from frozen *eligible-ids* manifests. Attention is eager for Qwen2-VL-7B and sdpa for SmolVLM2-2B, matching every committed legacy run.

**Training recipe (verbatim seed-0 recipe).** Each fresh seed is a General-LoRA run on the general training manifest (2000 rows; 95/5 split frozen at seed 42) with the seed-0 recipe: epochs 2, lr  $10^{-4}$ , LoRA  $r=8$ ,  $\alpha=16$ , dropout 0.05, bf16, gradient checkpointing. The only intentional difference from seed-0 is the explicit per-run seed (101/202/303) applied to torch/cuda/numpy at driver start. Three seeds per backbone were trained on A6000 GPUs.

**Fresh-seed analysis.** Seeds are independent training draws, not ordered stages; we report per-seed values, means, and standard deviations, and never “monotonic across seeds.” All comparisons are within-backbone; preprocessing is identical within a backbone.

**Independent numerical audit.** Every headline quantity in this paper was recomputed by a standalone script (`scripts/audit_paper2_numbers_independent.py`) that imports no analyzer module and reads only raw prediction CSVs/JSONL, frozen protocol manifests, and committed analysis JSONs as comparison targets. The audit hard-fails on any deterministic-proportion discrepancy above  $10^{-6}$ , any row-count/ID mismatch, and any terminology mismatch. It passed with zero issues, and the committed seed-0 tier-a values were reproduced byte-identically (7B normal 0.82414579; 2B normal 0.76492027; absolute difference 0.0).

## 4. Results

All values below are audit-sourced (Step-1 independent audit, PASS). Seeds are reported per-draw; means/SDs are over the three fresh seeds.

| family      | checkpoint | $\Delta A$ | $G$    | $\Delta G$ |
|-------------|------------|------------|--------|------------|
| Qwen2-VL-7B | seed-0     | +0.0542    | 0.3522 | +0.0456    |
| Qwen2-VL-7B | seedA      | +0.0547    | 0.3544 | +0.0478    |
| Qwen2-VL-7B | seedB      | +0.0456    | 0.3444 | +0.0378    |
| Qwen2-VL-7B | seedC      | +0.0515    | 0.3531 | +0.0465    |
| SmolVLM2-2B | seed-0     | +0.0287    | 0.2975 | +0.0305    |
| SmolVLM2-2B | seedA      | +0.0305    | 0.3007 | +0.0337    |
| SmolVLM2-2B | seedB      | +0.0323    | 0.2957 | +0.0287    |
| SmolVLM2-2B | seedC      | +0.0319    | 0.2993 | +0.0323    |
| Qwen3-VL-8B | tuned      | +0.0323    | 0.3827 | +0.0355    |

Table 1. Headline adaptation quantities.  $G = A_{\text{normal}} - A_{\text{shuffle}}$ ;  $\Delta G$  is relative to zero-shot. All values from the independent audit.

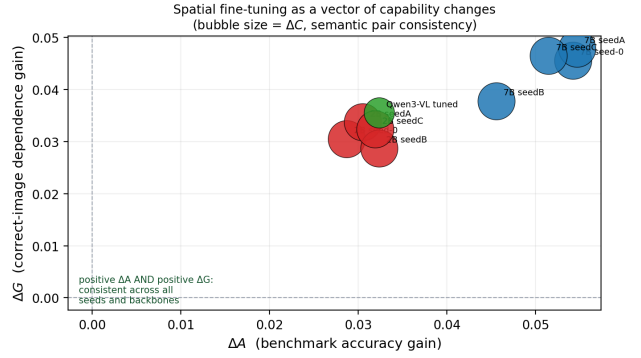


Figure 1. Conceptual decomposition:  $\Delta A$  (x) versus  $\Delta G$  (y) for all nine checkpoints (7B seed-0 + three seeds, 2B seed-0 + three seeds, Qwen3-VL-8B tuned); bubble size encodes  $\Delta C$  (semantic pair consistency change). All points lie in the positive-positive quadrant.

**Benchmark gain ( $\Delta A$ ) and correct-image dependence ( $\Delta G$ ).** Table 1 reports the per-seed quantities.  $\Delta A$  is positive for every seed on both backbones (7B fresh-seed mean  $+0.0506 \pm 0.0046$ ; 2B fresh-seed mean  $+0.0316 \pm 0.0009$ ). The normal-minus-shuffle gap  $G$  widens under tuning in every family, and the zero-shot-relative change  $\Delta G$  is positive for every seed (7B fresh-seed mean  $+0.0440 \pm 0.0054$ ; 2B fresh-seed mean  $+0.0316 \pm 0.0026$ ). Shuffle accuracy itself stays flat (e.g., 2B zero-shot 0.4692 vs. tuned 0.4674–0.4729), so the gain is in using the correct image, not in memorizing image-independent answers.

**Semantic pair consistency ( $\Delta C$ , relcomp).** Table 2 shows relcomp  $C_{\text{pair}}$ . Fresh seeds cluster tightly around seed-0 in both families (7B: 0.6547–0.6652 vs. 0.6772; 2B: 0.4985–0.5105 vs. 0.5015); the audit confirms every fresh seed is within 0.05 of the legacy General checkpoint. The facing-antonym diagnostic (facingcomp,  $n = 103$ ) shows the same pattern of tight seed clustering (2B seed-0 0.3469; fresh seeds 0.3429–0.3714).

| family      | checkpoint | $C_{\text{pair}}$ (relcomp) |
|-------------|------------|-----------------------------|
| Qwen2-VL-7B | seed-0     | 0.6772                      |
| Qwen2-VL-7B | seedA      | 0.6637                      |
| Qwen2-VL-7B | seedB      | 0.6547                      |
| Qwen2-VL-7B | seedC      | 0.6652                      |
| SmolVLM2-2B | seed-0     | 0.5015                      |
| SmolVLM2-2B | seedA      | 0.4985                      |
| SmolVLM2-2B | seedB      | 0.5045                      |
| SmolVLM2-2B | seedC      | 0.5105                      |

Table 2. Relcomp pair consistency ( $n = 666$ ). Fresh seeds stay within 0.05 of the legacy General checkpoint in both families.

| family      | checkpoint | $A_{\text{transform}}$ | $C_{\text{pair}}$ | both_correct |
|-------------|------------|------------------------|-------------------|--------------|
| Qwen2-VL-7B | zero-shot  | 0.6367                 | 0.6163            | 0.5388       |
| Qwen2-VL-7B | seed-0     | 0.6571                 | 0.6857            | 0.5959       |
| Qwen2-VL-7B | seedA      | 0.6490                 | 0.6490            | 0.5796       |
| Qwen2-VL-7B | seedB      | 0.6571                 | 0.6898            | 0.6041       |
| Qwen2-VL-7B | seedC      | 0.6449                 | 0.6653            | 0.5837       |
| SmolVLM2-2B | zero-shot  | 0.4980                 | 0.3184            | 0.2531       |
| SmolVLM2-2B | seed-0     | 0.5224                 | 0.3469            | 0.2980       |
| SmolVLM2-2B | seedA      | 0.5469                 | 0.3429            | 0.3061       |
| SmolVLM2-2B | seedB      | 0.5469                 | 0.3633            | 0.3143       |
| SmolVLM2-2B | seedC      | 0.5469                 | 0.3714            | 0.3224       |

Table 3. Tier-C hflip\_flip ( $n = 245$ ).  $A_{\text{transform}}$  is transformed-answer accuracy;  $C_{\text{pair}}$  is the answer-update rate  $P(\text{mirrored} \neq \text{normal})$ ; both\_correct is joint correctness. Three separate quantities, never collapsed.

**Transformation behavior under global reflection (Tier C).** Table 3 reports the three Tier-C quantities separately. Under global horizontal reflection (language fixed, left/right truth flips):

- **Transformed accuracy** ( $A_{\text{transform}}$ ): 2B rises from 0.4980 (zero-shot) to 0.5224 (seed-0) and 0.5469 for each fresh seed; 7B 0.6367  $\rightarrow$  0.6571 with fresh seeds 0.6490–0.6571.
- **Answer-update rate** ( $C_{\text{pair}}$ ): 2B 0.3184 (zero-shot)  $\rightarrow$  0.3469 (seed-0), with all three fresh seeds higher than zero-shot (0.3429/0.3633/0.3714); 7B 0.6163  $\rightarrow$  0.6857, fresh seeds 0.6490/0.6898/0.6653.
- **Joint correctness** (both\_correct): 2B 0.2531  $\rightarrow$  0.2980 (seed-0)  $\rightarrow$  0.3061/0.3143/0.3224.

The audit confirms all fresh 2B  $C_{\text{pair}}$  values exceed the 2B zero-shot value, and all fresh-seed  $C_{\text{pair}}$  values (both families) are within 0.05 of the legacy General checkpoint.

**Contemporary-backbone extension (Qwen3-VL-8B, post-confirmatory).** Table 1 includes the Qwen3-VL-8B row:  $\Delta A + 0.0323$ ,  $\Delta G + 0.0355$ , and hflip\_flip transformed accuracy  $+0.0449$  (0.6571  $\rightarrow$  0.7020). This extension is explicitly exploratory: it computed only  $\Delta A$ ,  $\Delta G$ , and

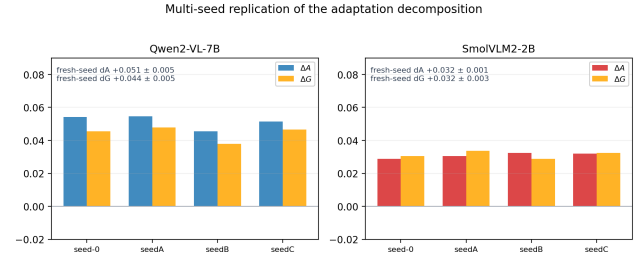


Figure 2. Multi-seed replication: per-seed  $\Delta A$  and  $\Delta G$  for both confirmatory families, with fresh-seed mean  $\pm$  SD annotations.

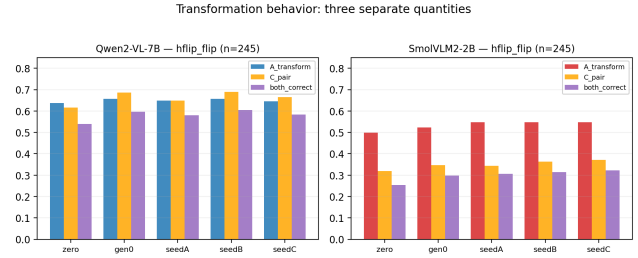


Figure 3. Transformation behavior:  $A_{\text{transform}}$ ,  $C_{\text{pair}}$ , and both\_correct per checkpoint, hflip\_flip ( $n = 245$ ), both families.

$A_{\text{transform}}$ :  $C_{\text{pair}}$  was not computed for it, and no response-law-compliance claim is made. It is labeled post-confirmatory external validation, not preregistered confirmatory evidence.

### Claim-level summary (all audited).

- $\Delta A > 0$  for all fresh seeds, both confirmatory backbones: PASS.
- $\Delta G > 0$  for all fresh seeds, both confirmatory backbones: PASS.
- Fresh-seed  $C_{\text{pair}}$  within 0.05 of legacy General (7B and 2B): PASS.
- All fresh 2B hflip\_flip  $C_{\text{pair}} > 2B$  zero-shot: PASS.
- Qwen3-VL-8B supports only the stated  $\Delta A/\Delta G/A_{\text{transform}}$  quantities: PASS.

## 5. Discussion and Limitations

**What the decomposition shows.** The headline result is that ordinary spatial fine-tuning is not a scalar capability update. Benchmark accuracy, correct-image dependence, and semantic pair consistency move together in direction (all positive for  $\Delta A$  and  $\Delta G$ ;  $\Delta C$  stable) but they are distinct quantities: shuffle accuracy stays flat while the gap widens; answer-update rates improve while transformed accuracy moves differently; and the three Tier-C quantities never collapse into one number. A model can get the mirrored item right without updating its answer relative to its own original prediction, and vice versa; reporting them separately is essential.



**Seed robustness.** Every component replicates across three fresh training draws per family. The fresh-seed standard deviations are small (7B  $\Delta A$  SD 0.0046; 2B  $\Delta A$  SD 0.0009), which closes the single-checkpoint-anecdote criticism that a training effect could be a lucky draw.

**Relationship to prior work.** We do not claim the normal-minus-shuffle gap as novel: VRS in Beyond Accuracy [6] is the same construction. We do not claim to reproduce VisualFLIP [7]: our reflection is a global image intervention, and we state the difference. Our novelty is the joint, seed-robust, cross-backbone decomposition of ordinary fine-tuning into  $\Delta A$ ,  $\Delta G$ ,  $\Delta C$ , and Tier-C behavior.

**Limitations.** (1) Two backbone families (7B and 2B) plus one contemporary extension; the extension computed only  $\Delta A/\Delta G/A_{\text{transform}}$  and is exploratory. (2) The Tier-C intervention is a global horizontal reflection; VisualFLIP-style minimal local edits were not run because the official dataset is not yet public. (3) All comparisons are within-backbone with identical preprocessing; we do not claim resolution or architecture independence. (4) The shuffle condition tests dependence on the *correct image*, which we call correct-image dependence, not “causal grounding”; stronger causal language is reserved for genuine counterfactual evidence. (5) A text-only control adapter was considered but not run; it remains future work. (6) Seeds are independent draws; we do not claim a “monotonic” seed trend anywhere.

## 6. Conclusion

Ordinary spatial LoRA fine-tuning on VSR produces a reproducible vector of capability changes, not a single benchmark gain. Across three fresh training seeds and two backbone families,  $\Delta A$  (benchmark accuracy) and  $\Delta G$  (correct-image dependence) are positive for every seed, semantic pair consistency  $\Delta C$  stays within 0.05 of the legacy General checkpoint, and Tier-C transformation behavior improves answer-update rates while remaining a set of three separately reported quantities. A contemporary Qwen3-VL-8B extension reproduces the primary  $\Delta A/\Delta G$  pattern and the transformed-accuracy gain. Every number was independently recomputed from raw artifacts and the frozen protocol; the full claim hierarchy, numerical audit, literature audit, figures, and tables are archived in the repository. We recommend that spatial fine-tuning be evaluated and reported as this joint vector—accuracy, correct-image dependence, semantic consistency, and transformation behavior—rather than collapsed into a single accuracy number.

## References

[1] S Divakar Bhat and Toshihiko Yamasaki. Consistent yet wrong: Evidence insensitivity in spatial vision-language

- models. *arXiv preprint arXiv:2606.02742*, 2026. 1, 2
- [2] Zixuan Lan, Luzhe Sun, Matthew R. Walter, and Jiawei Zhou. Seeing without looking: Do vision-language benchmarks really test vision? *arXiv preprint arXiv:2605.22903*, 2026. CVPR 2026 Workshop (GRAIL-V). 1, 2
- [3] Fangyu Liu, Guy Emerson, and Nigel Collier. Evaluating the impact of model scale for compositional generalization in visual grounding. In *EMNLP*, 2023. 1, 2
- [4] Yunhang Qian, Jiaquan Yu, Jiawei Liu, Meng Wang, Hongwei Bran Li, and Xiaobin Hu. Med-opd: Improving medical vision-language models via evidence-aware on-policy distillation. *arXiv preprint arXiv:2607.16303*, 2026. 2
- [5] Anas Zafar, Leema Krishna Murali, Siddhant Bharadwaj, Ashish Vashist, and Jia Wu. Do medical vision language models actually see? a counterfactual grounding framework and hard-negative contrastive training for visually-reliant medical vlms. *arXiv preprint arXiv:2607.03647*, 2026. 1, 2
- [6] Anas Zafar, Leema Krishna Murali, and Ashish Vashist. Beyond accuracy: Evaluating visual grounding in multimodal medical reasoning. *arXiv preprint arXiv:2603.03437*, 2026. 1, 2, 5
- [7] Didi Zhu, Changrui Chen, Stefanos Zafeiriou, and Jiankang Deng. Visualflip: Do predictions depend on task-critical visual evidence in multimodal reasoning? *arXiv preprint arXiv:2606.07872*, 2026. 1, 2, 5